Lab Exercise 16- Creating a ConfigMap from a Folder

In this lab exercise, you will learn how to create a ConfigMap in Kubernetes from a folder containing multiple files. This allows you to manage multiple configuration files as a single ConfigMap.

Step 1: Set Up Kubernetes Cluster

Ensure you have access to a Kubernetes cluster. You can use a local setup with Minikube, kind, or use a cloud-based Kubernetes service.

Step 2: Prepare the Folder with Multiple Files

Create a directory named configmap-folder:

```
mkdir configmap-folder
cd configmap-folder/
```

```
devanksilswal@devanks-MacBook-Air ex_s % mkdir configmap-folder
cd configmap-folder/
```

Create multiple configuration files inside the directory:

config1.txt with content:

```
key1=value1
```

config2.txt with content:

```
key2=value2
```

config3.txt with content:

```
key3=value3
```

```
devanksilswal@devanks-MacBook-Air configmap-folder % touch config1.txt
devanksilswal@devanks-MacBook-Air configmap-folder % nano config1.txt
devanksilswal@devanks-MacBook-Air configmap-folder % touch config2.txt
devanksilswal@devanks-MacBook-Air configmap-folder % nano config2.txt
devanksilswal@devanks-MacBook-Air configmap-folder % touch config3.txt
devanksilswal@devanks-MacBook-Air configmap-folder % nano config3.txt
```

Verify the directory structure:

```
ls configmap-folder/
cd ..
```

```
devanksilswal@devanks-MacBook-Air configmap-folder % ls
cd ..
config1.txt     config2.txt     config3.txt
```

You should see the files config1.txt, config2.txt, and config3.txt.

Step 3: Create a ConfigMap from the Folder

Create a ConfigMap named folder-configmap from the folder:

```
kubectl create configmap folder-configmap --from-file=configmap-folder/
```

```
devanksilswal@devanks-MacBook-Air ex_s % kubectl create configmap folder-configmap --from-file=configmap-folder/
configmap/folder-configmap created
```

Verify the ConfigMap:

```
kubectl get configmap folder-configmap -o yaml
```

```
devanksilswal@devanks-MacBook-Air ex_s % kubectl get configmap folder-configmap -o yaml
apiVersion: v1
data:
  config1.txt: |
    key1=value1
  config2.txt: |
    key2=value2
  config3.txt: |
    key3=value3
kind: ConfigMap
metadata:
  creationTimestamp: "2026-04-17T11:11:56Z"
  name: folder-configmap
  namespace: my-namespace
  resourceVersion: "179823"
  uid: ef1ab252-893a-4ecc-a301-6300aceafee1
```

You should see the folder-configmap with the contents of all the files in the configmap-folder.

Step 4: Use the ConfigMap in a Pod

Create a file named **pod-folder-configmap.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: configmap-pod
spec:
  containers:
  - name: nginx
    image: nginx
    envFrom:
    - configMapRef:
        name: folder-configmap
```

In this manifest:

- The Pod mounts the folder-configmap at /etc/config.
- The busybox container concatenates and prints the contents of the files.

Apply the manifest to create the Pod:

```
kubectl apply -f pod-folder-configmap.yaml
```

```
devanksilswal@devanks-MacBook-Air ex_s % touch pod-folder-configmap.yaml
devanksilswal@devanks-MacBook-Air ex_s % nano pod-folder-configmap.yaml
devanksilswal@devanks-MacBook-Air ex_s % kubectl apply -f pod-folder-configmap.yaml
pod/configmap-pod created
```

Check the status of the Pod:

```
kubectl get pods
```

```
devanksilswal@devanks-MacBook-Air ex_s % kubectl get pods
NAME            READY   STATUS    RESTARTS   AGE
configmap-pod   1/1     Running   0          6s
```

View the Pod logs to see the content of the mounted files:

```
kubectl exec -it configmap-pod -- printenv
```

```
devanksilswal@devanks-MacBook-Air ex_s % kubectl exec -it configmap-pod -- printenv
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=configmap-pod
TERM=xterm
config2.txt=key2=value2

config3.txt=key3=value3

config1.txt=key1=value1

KUBERNETES_SERVICE_HOST=10.96.0.1
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PORT=443
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
NGINX_VERSION=1.29.8
NJS_VERSION=0.9.6
NJS_RELEASE=1~trixie
ACME_VERSION=0.3.1
PKG_RELEASE=1~trixie
DYNPKG_RELEASE=1~trixie
HOME=/root
```

You should see the contents of config1.txt, config2.txt, and config3.txt printed.

Step 5: Clean Up

After completing the exercise, clean up the resources created:

```
kubectl delete pod folder-configmap-pod
kubectl delete configmap folder-configmap
rm -r configmap-folder
```

```
devanksilswal@devanks-MacBook-Air ex_s % kubectl delete pod folder-configmap-pod
kubectl delete configmap folder-configmap
rm -r configmap-folder
Error from server (NotFound): pods "folder-configmap-pod" not found
configmap "folder-configmap" deleted from my-namespace namespace
devanksilswal@devanks-MacBook-Air ex_s % kubectl delete pod configmap-pod
pod "configmap-pod" deleted from my-namespace namespace
```